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Unprecedented sensor network in rewetted fens in northeastern Germany reveals high variability of hydrological conditions

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We established a sensor network to monitor basic environmental state variables in rewetted temperate fens at an unprecedented scale as part of the TRR 410 WETSCAPES2.0 project (funded by DFG, Project-ID 531801029). Monitoring locations were selected from a pool of all peatland rewetting projects in the federal state of Mecklenburg-Vorpommern in northeastern Germany using a standardized, remote-sensing-based procedure to obtain the most representative homogeneous area in each site. At each location, we installed sensors that measure groundwater table, soil moisture, and soil/air temperature and transmit their data live via LoRaWAN.

Over the last months, more than 80 sites have been set up in coastal and freshwater fens. Despite all sites being considered “rewetted”, they cover a wide hydrological gradient: In the first few months after setup, the average water levels in fall and winter ranged from more than 50 cm below ground surface up to almost 70 cm above ground. Also, already the first months of data indicate large differences in water level amplitude per site, ranging from less than 3 cm to more than 60 cm. No relationship between mean water level and amplitude of water level fluctuations could be observed. Several sites consistently showed soil water content in the upper layer less than 75%.

As the data set continues to grow, our data will help to enhance functional understanding of spatio-temporal implications of peatland rewetting, as well as serve in practical planning of future rewetting projects.